Department of Engineering Design,
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ED5235 :
Power Electronics and Motor Drives for Electrified Vehicles

Project Report

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1 Introduction

1.1 Question

Toyota engineers are planning to test the proposed converter for a hybrid electric vehicle that requires a 1-kW bidirectional isolated converter for an auxiliary power module with 12 V and 48 V from a 500 V DC link (HV side). For the operation, the switching frequency chosen by the design engineer is 10 kHz.

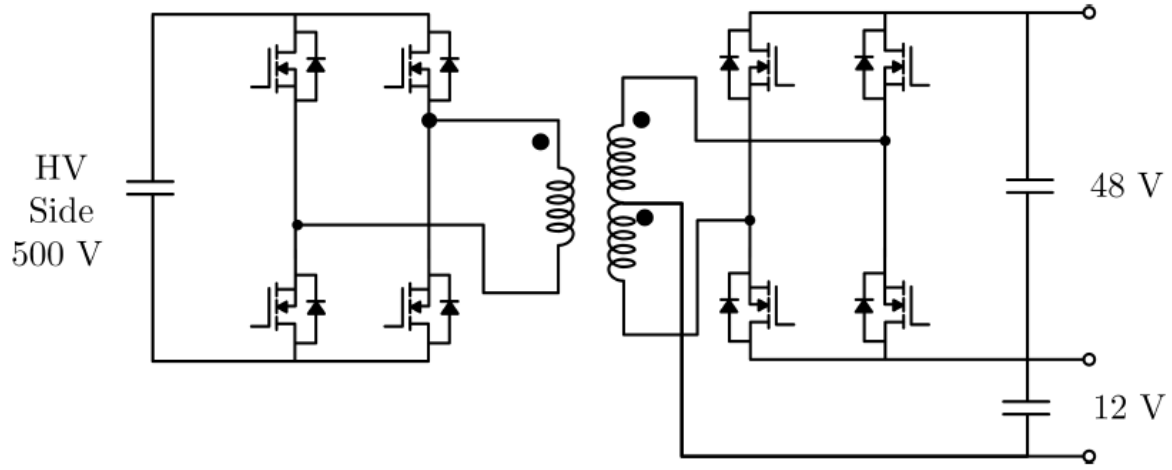


Figure 1: Proposed Converter

1. Determine the component parameters to have a maximum allowable inductor current ripple of 10 % and 0.5% voltage ripple at output.
2. Select the components for the converter based on the datasheets
3. Corresponding to the component selected, calculate the losses and calculate the analytical efficiency of the power converter.
4. Simulate the converter in MATLAB or any other suitable software and report the required waveforms.
5. From the simulation results, calculate the power converter's efficiency based on the datasheet for the selected component.

1.2 Circuit Analysis

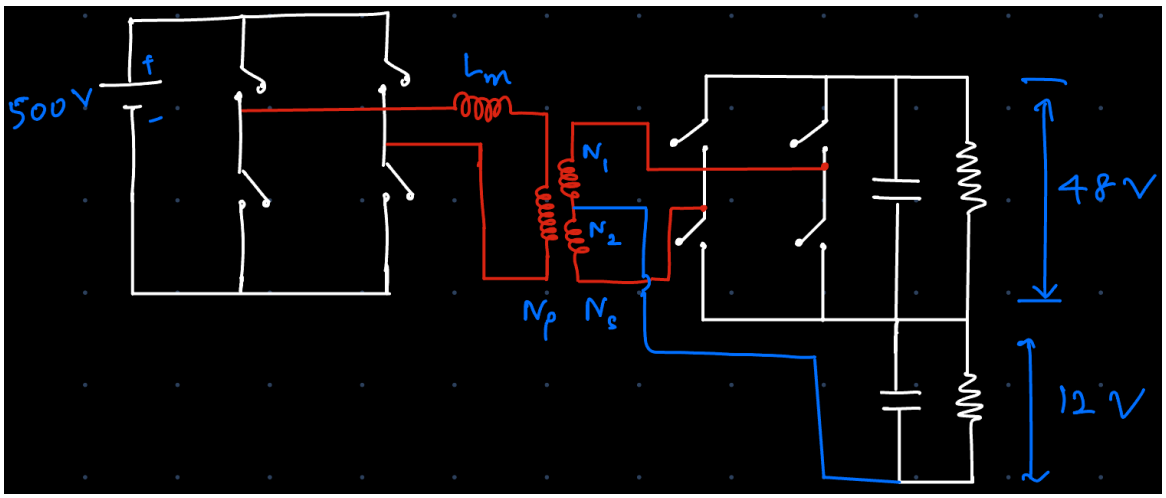


Figure 2: Desired Converter

The converter topology shown above features four MOSFET switches. For analytical clarity, the transformer is modeled by separating its ideal and leakage components, resulting in an explicit leakage inductance (L_m) in the circuit. The design also incorporates two DC-link capacitors and their respective load branches.

To streamline the analysis, the **first iteration** of the converter has omitted the 12V output while **retaining** the three-winding transformer configuration. This simplification yields the following circuit:

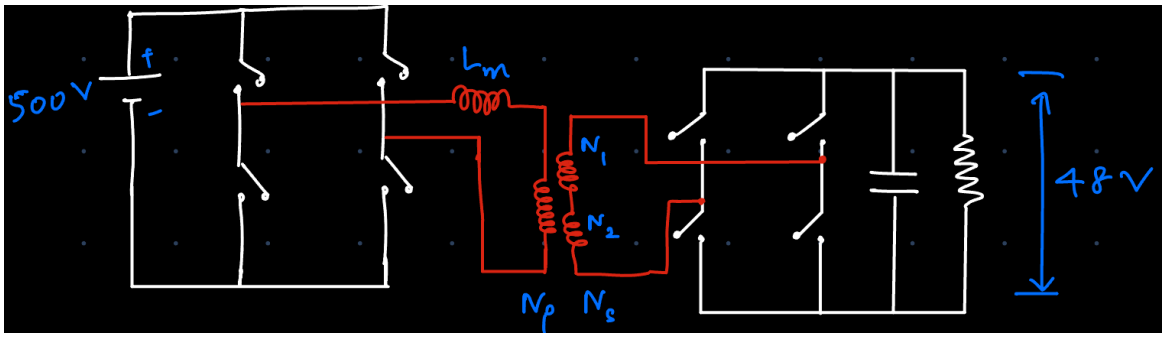


Figure 3: Simplified Converter

1.3 Dual-Active Bridge Converter

The circuit depicted above represents a Dual Active Bridge (DAB) converter, which is widely used for high-frequency, isolated DC-DC power conversion. The DAB operates as a phase-shifted, full-bridge converter, enabling both bidirectional power flow and galvanic isolation.

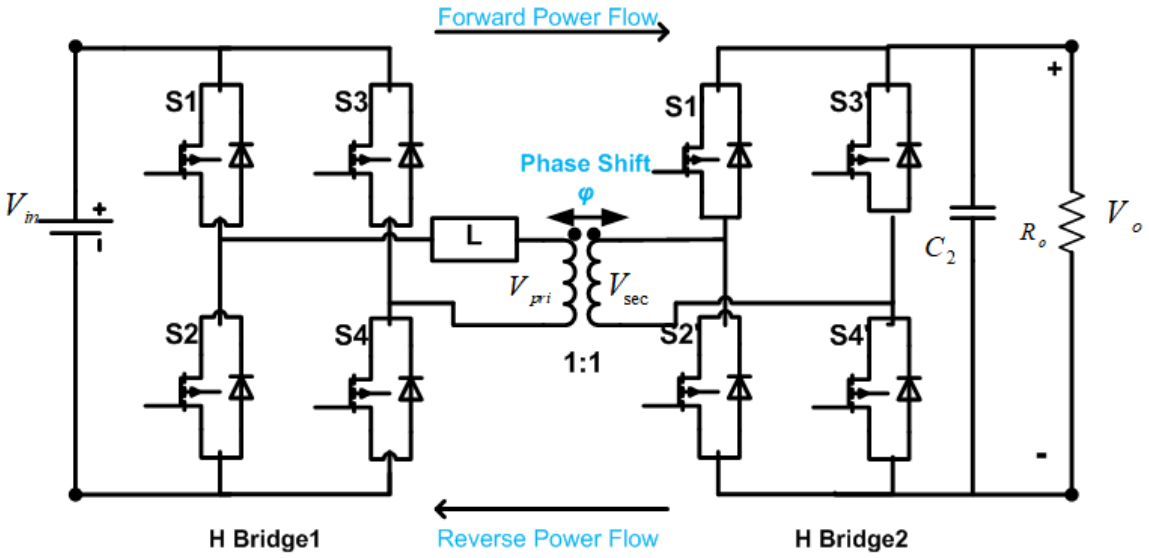


Figure 4: Dual Active Bridge Converter

- **Topology:** Consists of two full-bridge inverter stages (primary and secondary), each formed by four active switches (typically MOSFETs or IGBTs).
- **Isolation:** The bridges are interconnected via a high-frequency transformer, providing galvanic isolation and voltage scaling between input and output.
- **Leakage Inductance:** Incorporates leakage inductance (either inherent to the transformer or added externally) to facilitate energy transfer and smooth current waveforms.
- **DC-Link Capacitors:** Utilizes DC-link capacitors on both input and output sides to stabilize voltage and filter out voltage ripple.
- **Control:** Power transfer is regulated through the phase shift between the gate signals of the primary and secondary bridges.
- **Bidirectional Operation:** Supports bidirectional power flow, making it suitable for applications such as battery energy storage, electric vehicle charging, and renewable energy systems.
- **Soft Switching:** Enables efficient soft-switching operation (ZVS/ZCS) over a wide range of load conditions through appropriate phase shift control and transformer parameter selection.

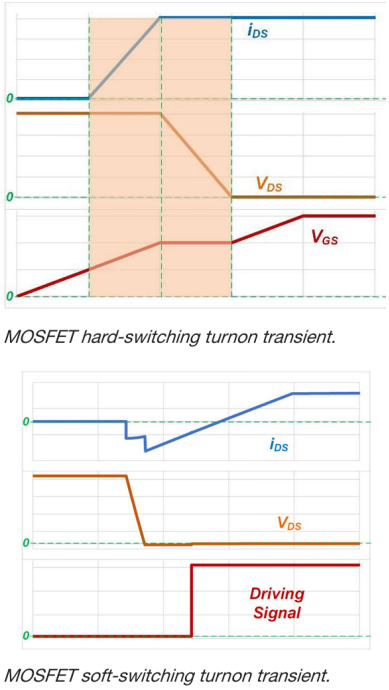


Figure 5: Switching Comparison

- **Scalability:** The topology is scalable and can be adapted for multi-port or multi-level configurations for advanced power electronic systems.

1.4 Switching Sequence

The switching sequence is divided into four intervals based on the inductor current waveform and phase shift between the voltages at the primary and secondary of the transformer.

During interval one, the inductor current waveform is both positive and negative, and hence, the current commutation follows the scheme shown in Figure 2-4 and Figure 2-5. During this interval, switches Q1 and Q4 in the primary and switches Q6 and Q7 in the secondary conduct current.

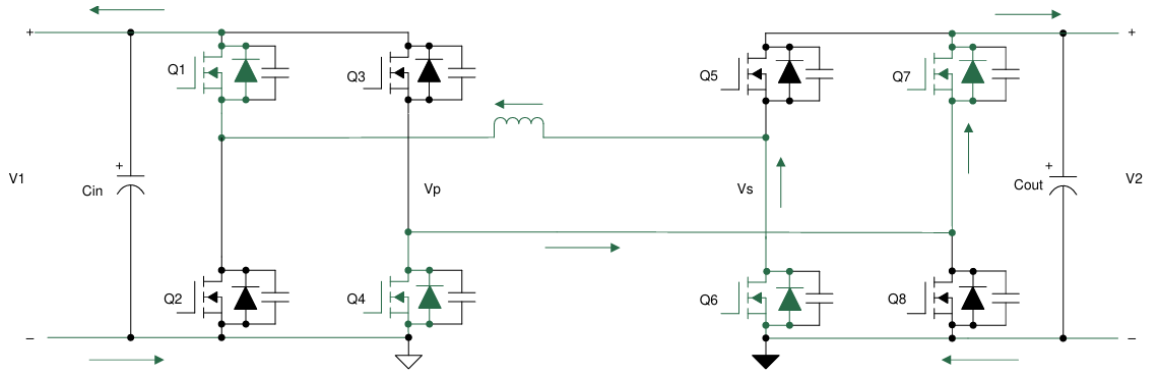


Figure 6: Interval 1 - Negative Inductor Current

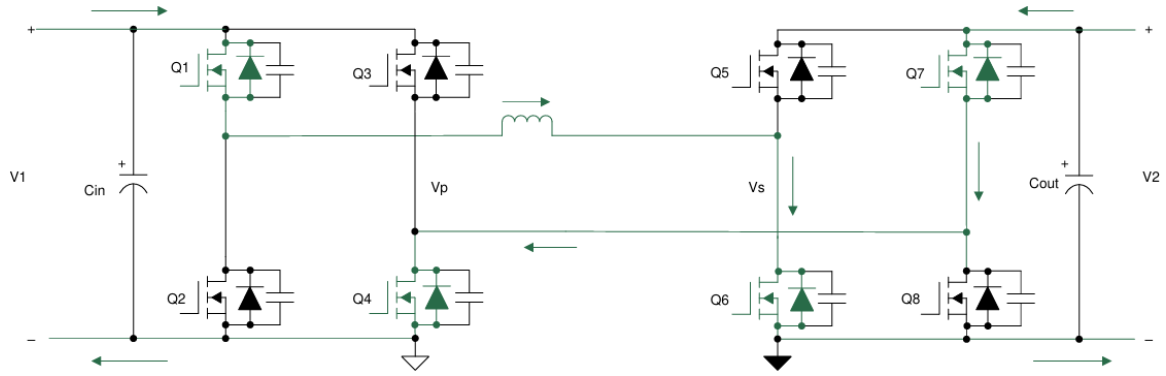


Figure 7: Interval 1 - Positive Inductor Current

During this interval, the voltage across the primary, Vp, is equal to V1, and the voltage across the secondary, Vs, is equal to V2. The difference between these voltages appears across the leakage inductor, and the slope of the current during this interval can be approximated as :

$$\frac{di}{dt} = \frac{V_1 + V_2}{L} \tag{1}$$

During interval two, the inductor current is positive. The voltage across the transformer primary is positive and is equal to V1, and the voltage across the secondary winding is positive and is equal to V2. Hence, the difference of these two voltages appears across the leakage inductor, and the slope of the rising current during this interval can be approximated as

$$\frac{di}{dt} = \frac{V_1 - V_2}{L} \tag{2}$$

During this interval, switches Q1 and Q4 remain turned on, but as the voltage across the secondary is now V2 with the inductor current positive, switches Q5 and Q8 turn on to conduct current. There is a small dead time period between the turn off of Q6 and Q7 and the turn on of Q5 and Q8. During this dead time, the phenomenon of **zero voltage switching** (ZVS) occurs.

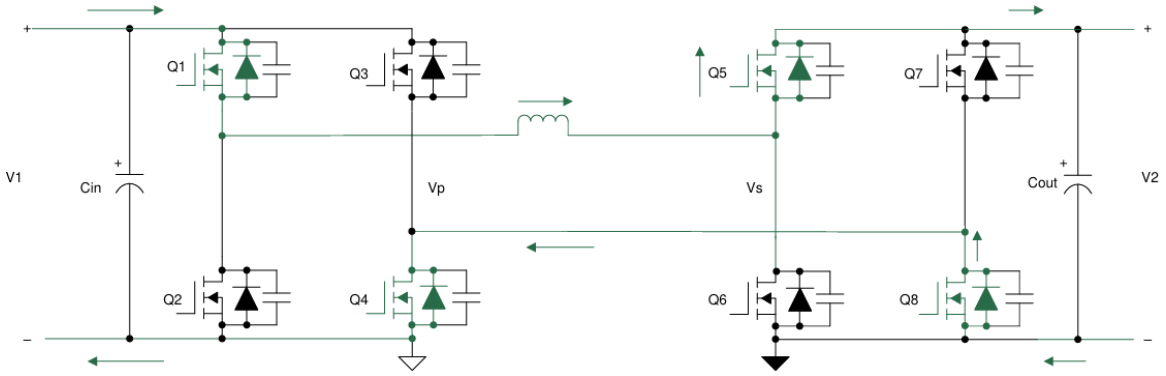


Figure 8: Interval 2

For interval 3, since it is a bidirectional Converter, there exists symmetry in the waveform, so from simple logic

$$\frac{di}{dt} = -\frac{V_1 + V_2}{L} \tag{3}$$

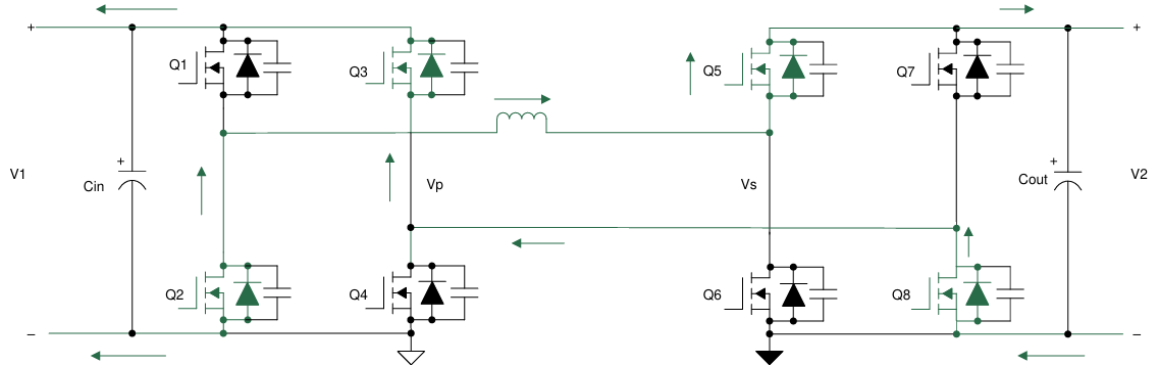


Figure 9: Interval 3 - Positive Inductor Current

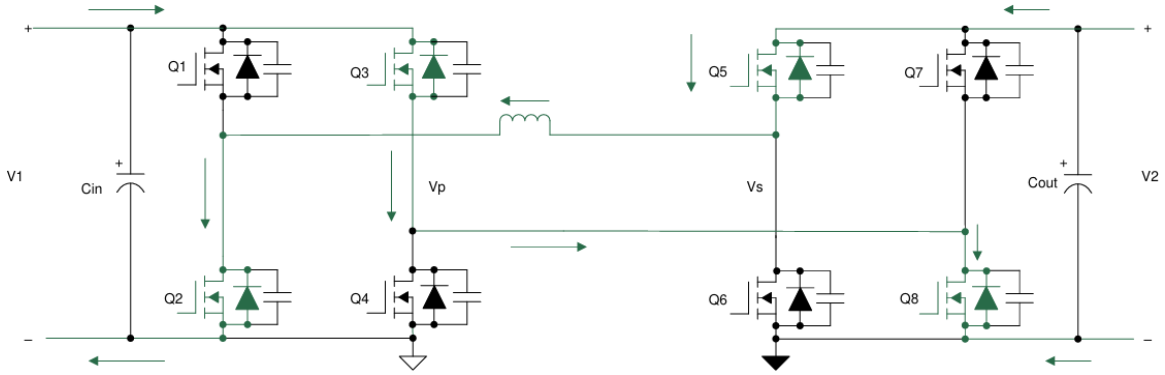


Figure 10: Interval 3 - Negative Inductor Current

Similarly for interval 4 :

$$\frac{di}{dt} = -\frac{V_1 - V_2}{L} \tag{4}$$

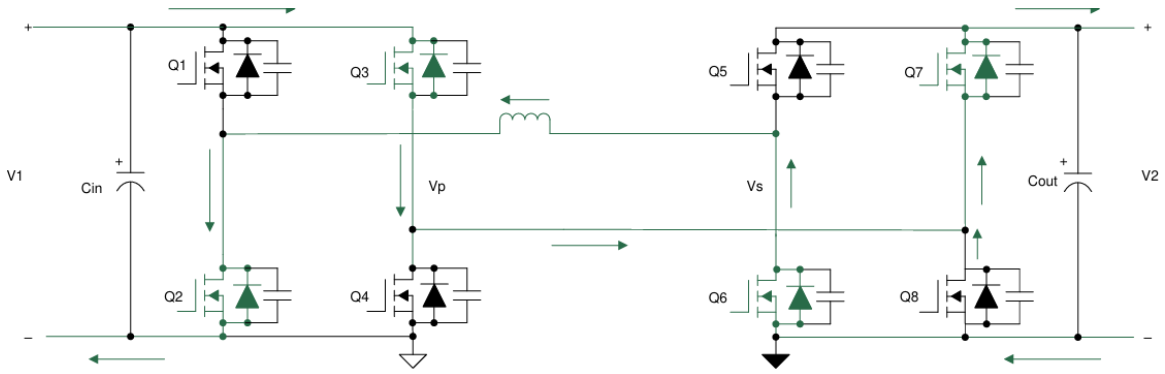


Figure 11: Interval 4

As discussed in the previous sections, the below waveforms exhibit a phase shift between the primary and secondary bridges, which directly influences the voltage across the inductor and shapes the inductor current profile, enabling controlled bidirectional power transfer in the converter.

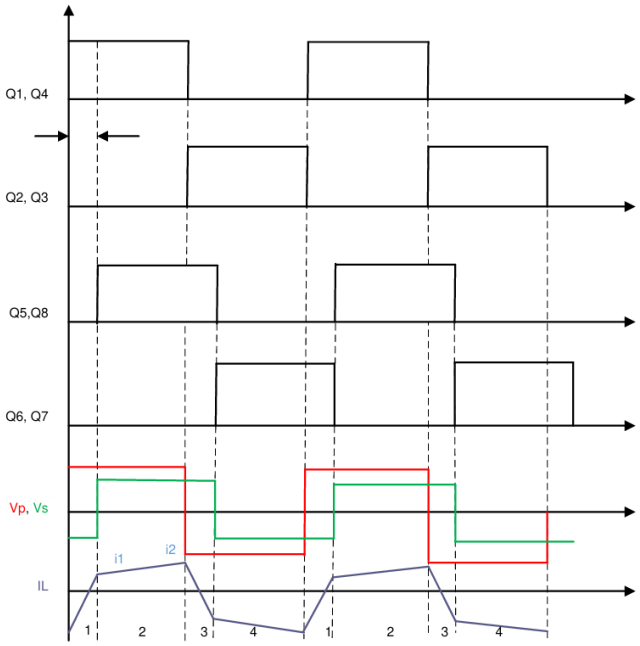


Figure 12: Waveforms

2 Component Analysis

They are many critical factors that influence the design of the power stage of a Dual Active bridge converter. The important ones include the choice of leakage inductor, desired operational phase shift, output capacitor specification, and switching operation frequency, choice of SiC MOSFETs, transformer, and desired ZVS operational range.
Many of these design parameters are interconnected, and the selection of any of these factors has a direct impact on other factors.

2.1 Transfomer Turn Ratio

In order to optimize the operation of the high-frequency transformer, the transformer ratio is obtained from the primary and secondary nominal voltages.

$$N_1 = \frac{V_{in,nom}}{V_{out1,nom}} = \frac{500}{48} \approx 10.42 \tag{5}$$

$$N_1 = \frac{V_{in,nom}}{V_{out2,nom}} = \frac{500}{12} \approx 41.67 \tag{6}$$

2.2 Leakage Inductor

The main specifications for creating a power converter system are input voltage V_1, output voltage V_2, and maximum power transfer. The power transfer relation for DAB is given by Equation 7. The best value for N is $V_1, nom/V_2, nom$

$$P = \frac{NV_1V_2\varphi(\pi - |\varphi|)}{2\pi^2 F_s L} \tag{7}$$

where

- N is the primary voltage
- V_1 is the primary side voltage
- V_2 is the secondary side voltage
- φ is the phase shift in radians
- F_s is the switching frequency
- L is the leakage inductance

Equation 7 shows that power transfer is maximum when $\varphi = \pi/2$.
Substituting φ as $\pi/2$ in the equation 7 yields

$$L = \frac{N \times V_1 \times V_2}{8 \times F_s \times P} \tag{8}$$

For $V_2 = 48V$ and $N = 10.42$

$$L_{max} = \frac{10.42 \times 500 \times 48}{8 \times 10^4 \times 1000} \approx 3.126mH \tag{9}$$

For $V_2 = 12V$ and $N = 41.67$

$$L_{max} = \frac{41.67 \times 500 \times 12}{8 \times 10^4 \times 1000} \approx 3.126mH \tag{10}$$

With V_1, V_2 and the required output power fixed, there are two output variables left to design. These are the leakage inductance L and the phase shift φ. With F_s set to $10kHz$, L is selected at 500μH at phase shift $\varphi = 0.130$. This allows a power transfer of ≈ 999.32 watts

The current at points i_1 and i_2 can be derived from 29 and 12

$$i_1 = 0.5 \times (2 \times \varphi - (1 - d) \times \pi) \times I_b \tag{11}$$

$$i_2 = 0.5 \times (2 \times d \times \varphi - (1 - d) \times \pi) \times I_b \tag{12}$$

where

- d is the voltage transfer ratio is given by equation 13
- I_{base} is the nominal base current is given by equation 14

$$d = N \times \frac{V_2}{V_1} \tag{13}$$

$$I_{base} = \frac{V_1}{\omega L} \tag{14}$$

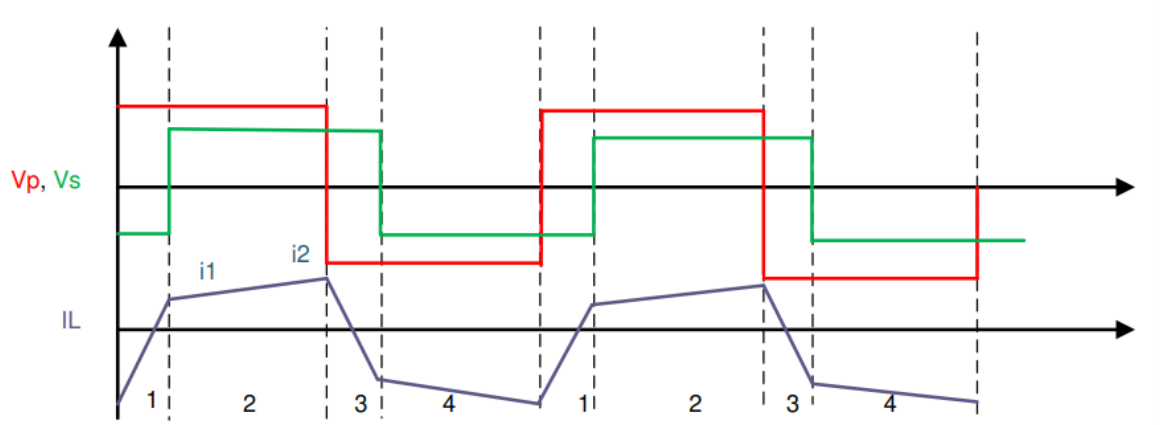


Figure 13: Inductor Current Waveform

Using Equation13 we calculate d_1 and d_2 for individual output voltages $V_2 = 48V$ and $V_2 = 12V$

$$d_1 = 10.42 \times \frac{48}{500} \approx 1 \tag{15}$$

$$d_2 = 41.67 \times \frac{12}{500} \approx 1 \tag{16}$$

Using Equation 14 we calculate I_{base}

$$I_{base} = \frac{500}{2 \times \pi \times 10^4 \times 500 \times 10^{-6}} \approx 15.92 \tag{17}$$

Calculating i_1 and i_2 for these output leads to:

$$i_1 = i_2 = 0.5 \times (2 \times 0.131) \times 15.92 \approx 2.069 \tag{18}$$

Note: $i_1 = i_2$ is only true for the nominal output voltage $V_2 = V_1/N$

2.3 Phase Shift

Phase shift is dependent on the value of leakage inductor. The phase shift for the required power is given by Equation 19

$$\varphi = \frac{\pi}{2} \times (1 - \sqrt{1 - \frac{8 \times F_s \times L \times P}{N \times V_1 \times V_2}} \tag{19}$$

For

- $F_s = 10^4 kHz$
- $L = 500\mu H$
- $P = 1kW$
- $N = 10.42$

- $V_1 = 500V$
- $V_2 = 48V$

Note: Product of $N \times V_1 \times V_2$ is the same for output voltages $V_2 = 48V$ and $V_2 = 12V$

Substituting these values in Equation 19 yields phase shift as $\varphi = 0.131$

Note: a high phase shift requires a high leakage inductance for power transfer. Using a high inductor leads to increased RMS currents in the primary and secondary side, which **affects the efficiency of the converter.**

3 Component Selection

3.1 Capacitor Selection

— The current that comes from the transformer secondary side is not a pure DC current; instead, it is a pulsating waveform called I_{HB2}. However, the load connected at the output (such as a battery or a motor) requires a steady DC current, denoted as I_{out} To maintain this steady output, the output capacitor must absorb or deliver the difference between the pulsating transformer current and the constant load current. This difference is defined as the capacitor current I_{cap}.

$$I_{cap} = I_{BH2} - I_{out} \tag{20}$$

Once the capacitor current I_{cap} is known, the next step is to understand how much charge the capacitor must store and release during each switching cycle. Since current is the rate of flow of charge, integrating I_{cap} over time gives the total charge accumulated in the capacitor.

To design the capacitor correctly, it is important to find the maximum amount of charge swing. This is done by subtracting the minimum value from its maximum value. This difference, called ΔQ, represents the total charge the capacitor must handle within one switching cycle:

$$\Delta Q = max(Q) - min(Q) \tag{21}$$

Finally, once ΔQ is known, the required output capacitance can be calculated. The relation between charge, capacitance, and voltage ripple is given by:

$$C_{out} = \frac{\Delta Q}{V_{ripple}} \tag{22}$$

Vripple is the maximum allowable output voltage ripple. A larger ΔQ or a tighter ripple specification will demand a bigger capacitor.

From the question, the maximum allowable inductor current ripple is 10% and 0.5% voltage ripple at output.

Using Equation 22 and a voltage ripple of 5% leads to required capacitance of $600\mu C$.These are the best values assuming no parasitics in the capacitors. In the current design Aluminum Electrolytic Capacitors with relatively high ESR are used. Therefore $1500\mu C$ of output capacitance is necessary to reduce the ripple to 5%.

3.2 SiC MOSFET Selection

To ensure reliable operation, the primary and secondary power stage switches must be capable of blocking the full input and output DC voltages. Silicon carbide (SiC) MOSFETs were selected for this application due to their superior performance compared to traditional silicon (Si) devices. SiC switches offer faster switching speeds, which contribute to significantly lower switching losses. Additionally, their much lower reverse recovery charge minimizes voltage and current overshoot, making them particularly well-suited for Dual Active Bridge (DAB) converters. The inherently lower on-state resistance of SiC devices also leads to reduced conduction losses during operation. Furthermore, SiC MOSFETs are capable of blocking higher voltages without breakdown, ensuring robust and efficient performance in high-voltage applications.

For our Design, on primary side we chose **SCT4036KR** with on-side resistance of $35m\Omega$ and on secondary side, we chose **Infineon IAUS300N08S5N012T** with on-side resitance $1.2m\Omega$. This was selected on safety factor of 1.5 times.

3.3 Transformer Selection

In this design, the transformer plays a critical role in determining the overall size, efficiency, and performance of the dual-output auxiliary converter. To meet the requirements of high efficiency, high power density, and compact size, careful selection and design of the transformer were necessary.

Increasing the operating frequency can significantly reduce the size of the magnetic components; however, excessive switching frequency can lead to increased losses due to skin and proximity effects. At 10 kHz, these high-frequency effects are manageable but must still be considered. The skin effect reduces the effective cross-sectional area for current conduction, increasing resistance at higher frequencies, while the proximity effect causes additional current crowding, further elevating AC losses. To mitigate these phenomena, conductor sizing and winding layouts must be optimized.

Given the need for tight control over leakage inductance to facilitate Zero Voltage Switching (ZVS) and to support efficient power transfer between the primary and secondary sides, a **planar transformer** architecture was selected for this design. Planar transformers offer several distinct advantages, including:

- High power density and compact form factor, reducing overall system volume.
- Superior thermal management due to their flat geometry and greater surface area.
- Greater flexibility in winding interleaving, minimizing AC resistance and proximity effect losses.
- Consistent parasitic elements, allowing precise control of leakage inductance and inter-winding capacitance.
- The ability to integrate additional magnetics (such as shim inductors) within the transformer structure, eliminating the need for external components.

Table 1: DC/DC Converter Electrical parameters

Parameters	Specifications
Phase Shift	-0.130 ¡ ¡ 0.130 (rad)
Total Leakage Inductance	500 µH
Turns Ratio (for 48V)	1: 10.42
Turns Ratio (for 12V)	1:41.67
Load resistance	2.304Ω
Input Voltage	500V
Output voltage1	48V
Output voltage2	12V
Input Current	2A
Output Current1	16.67A (800W split)
Output Current2	4.17A (200W split)
Power	1000W

In this converter, two secondary windings are used to provide isolated 48 V and 12 V outputs. The turns ratios are designed based on the high-side input voltage of 500 V, resulting in a primary-to-secondary ratio of approximately 10.42:1 for the 48 V output and 41.67:1 for the 12 V output. Each winding was individually optimized to provide the necessary isolation, voltage scaling, and current handling capacity.

The leakage inductance was intentionally engineered to support ZVS operation across a significant portion of the load range. However, achieving full soft switching at very light loads using leakage inductance alone would have required impractically high inductance values, leading to elevated RMS currents and reduced efficiency. Therefore, the leakage inductance was selected to enable ZVS up to approximately one-half to one-third of the rated load. At lighter loads, some hard switching is acceptable, as conduction losses decrease proportionally.

4 Loss Analysis

This section reviews the theoretical efficiency numbers obtained in the dual-active bridge. To arrive at the losses in different elements, the average and the RMS currents across the primary and secondary side are calculated. Details on the actual derivation of equations are out of scope for this design. The maximum power transfer in a dual-active bridge occurs at a phase shift of 90°. However, a high phase shift requires a high leakage inductance for power transfer. Using a high inductor leads to increased RMS currents in the primary and secondary side, which affects the efficiency of the converter.

4.1 SiC MOSFET and Diode Losses

As SiC is used in the power stage, the body diodes conduct only during the dead time, causing ZVS. In all other instances, the channel of SiC is turned on to conduct current.

- $F_s = 10^4 kHz$
- $L = 500\mu H$

- $P = 1kW$
- $\varphi = 0.130$
- $N = 10.42$
- $V_1 = 500V$
- $V_2 = 48V$

Calculating i_1 and i_2 for these inputs leads to $i_1 = i_2$ =2.069 A. $i_1 = i_2$ is only true for the nominal output voltage $V_1 = V_2 \times N$.

The RMS value can be calculated by Equation 23. Subsitituting in the equation yields $2.642A$

$$I_{switch,prim.rms} = \sqrt{\frac{1}{6} \times (i_1^2 + i_2^2 + (1 - \frac{2\varphi}{\pi}) \times i_1 \times i_2)} \tag{23}$$

The diode conducts for only a small fraction of time during the switching period, as in the dead time causing ZVS. The dead time chosen for this application is 200 ns.

$$I_{diode,prim} = i_2 \times \frac{t_{dead}}{T_S} = 0.00418A \tag{24}$$

The value of drain-source resistance corresponding to the applied gate voltage waveform is obtained from the SiC MOSFETs data sheet. This value is 36 mΩ. The forward voltage drop across the body diode is 3.3 V. The conduction losses across the four primary side FETs is calculated using Equation 25

$$P_{cond,prim} = 4 \times \left(I_{swit,prim,rms}^2 \times R_{ds,on} + I_{diode,prim} \times V_{fd,prim}\right) \approx 1.0588A \tag{25}$$

Similarly, the conduction losses are calculated across the secondary side FETs by scaling the primary side RMS currents with transformer turns ratio using Equation 26 and Equation 27. The on state resistance of the secondary side MOSFET is 1.3 mΩ. The forward voltage drop across the body diode is 0.9 V.

$$I_{swit,sec,rms} = N \times I_{swit,prim,rms} \approx 27.53A \tag{26}$$

$$I_{diode,sec} = N \times I_{diode,prim} \approx 0.044A \tag{27}$$

$$P_{cond,sec} = 4 \times \left(I_{swit,sec,rms}^2 \times R_{ds,on} + I_{diode,sec} \times V_{fd,prim}\right) \approx 4.1W \tag{28}$$

4.2 Switching Loss

The switching loss curves from the manufacturer are used to calculate switching losses.

Because the FETs turn on at zero voltage, only the turn-off loss coefficients are used for calculating the switching losses. Using the SCT4036KR data sheet information, a turn-off energy of $26\mu J$ is estimated for this operating condition. Which leads to $50W$ on the primary side.

$$P_{sw,turnoff,sec} = F_s \times E_{off} = 2.6W \tag{29}$$

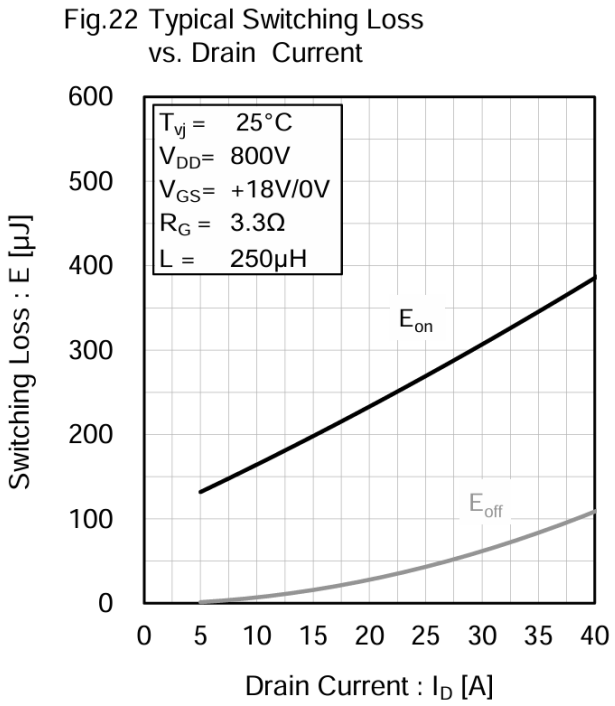


Figure 14: Typical Switching Loss vs. Drain Current

Table 2: Analytical Efficiency

Type of Loss	Loss (Watts)
SiC cinduction loss in primary and secondary side	5.1 W
SiC turn off switching loss	2.6 W
Trabsformer loss (estimated)	7 W
Gate driver loss + shunt resistor loss (estimated)	2 W
Efficiency	98.3 %

Efficiency (from simulation)

$$\eta = 1 - \frac{P_{loss}}{P_{in}} = 1 - \frac{27.5}{1000} = 97.5\% \tag{30}$$

5 Simulations

5.1 Simulink Model

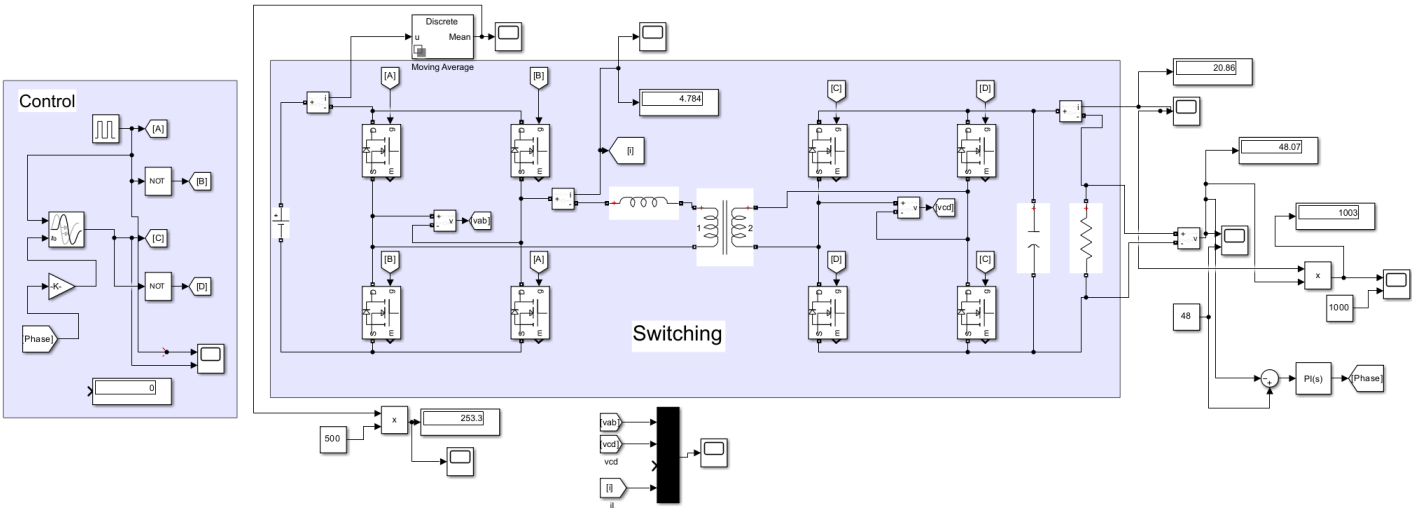


Figure 15: Simulink Model

Initial simulations were conducted using the simplified circuit configuration (Fig. 3) to establish stable operational parameters before transitioning to the desired converter topology. The analysis began with an open-loop control scheme for its straightforward implementation, which was later refined into a closed-loop architecture to enhance system efficiency and dynamic response. The final design iteration will incorporate the 12 V center-tapped output stage, which remains pending in the current implementation

5.2 Graphs

We first simulated an open loop switching sequence by calculating the φ, and then using it to phase shift the gate signals.

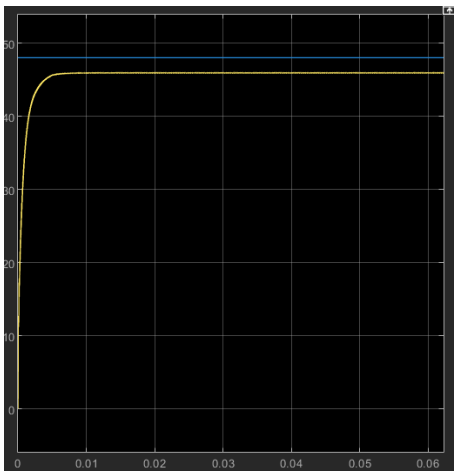


Figure 16: Voltage in open loop

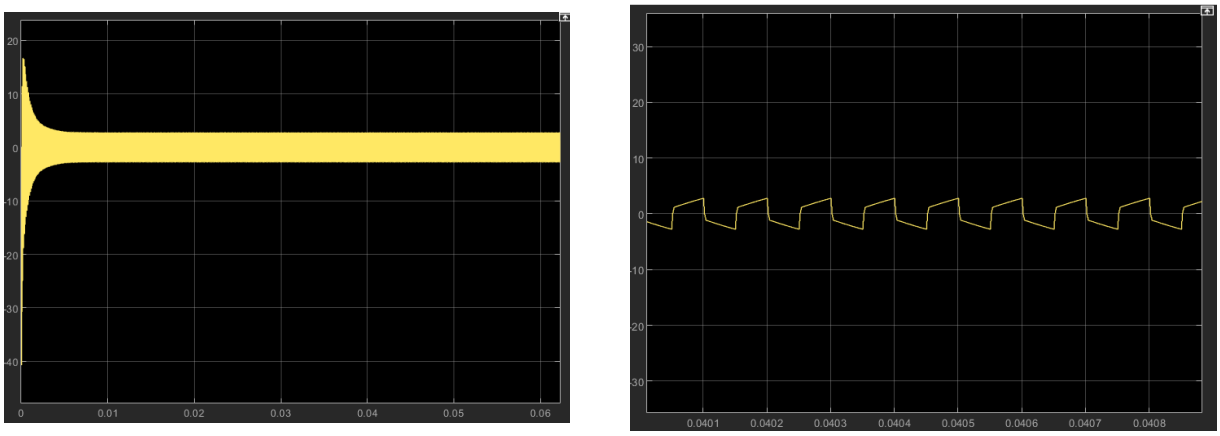


Figure 17: Inductor Current in open loop

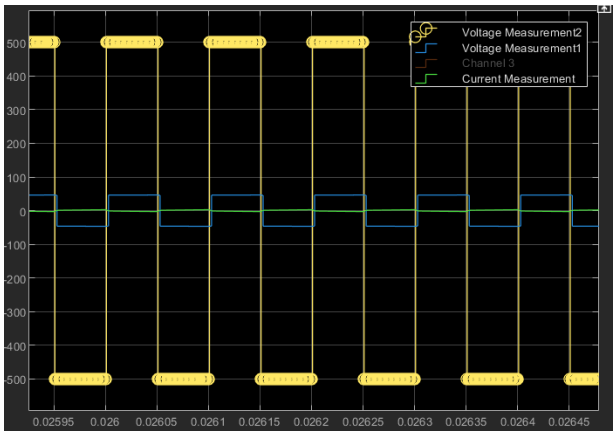


Figure 18: Transformer Voltages

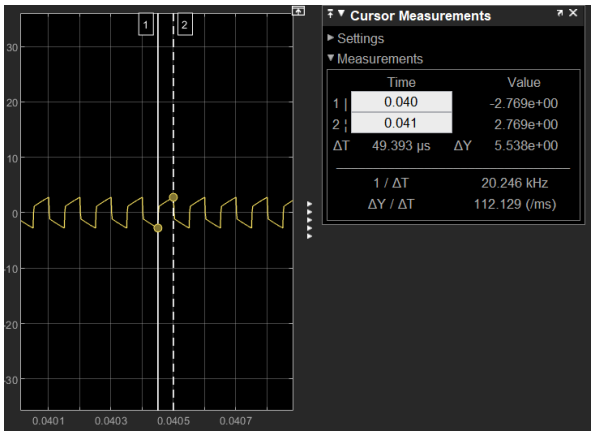


Figure 19: Inductor Current Ripple

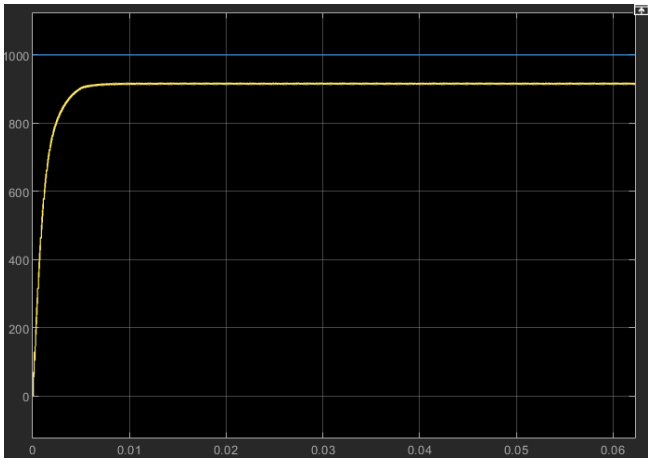


Figure 20: Power out in open loop

Since the open-loop configuration yielded inaccurate results, with the output voltage failing to approach the desired 48V, it became necessary to implement a closed-loop control system.

Closed-loop operation is essential for compensating bias and parameter variations across different system components, which can alter the phase shift (φ) and significantly affect overall converter performance.

Closed Loop Model :

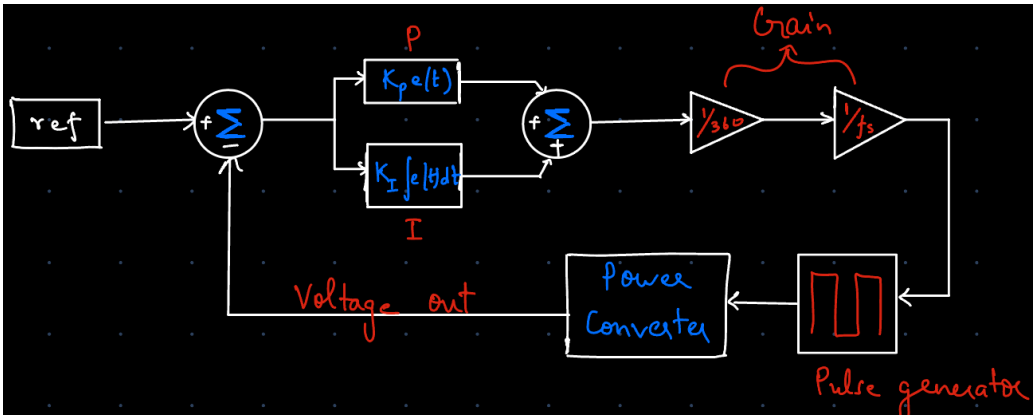


Figure 21: Closed Loop Model

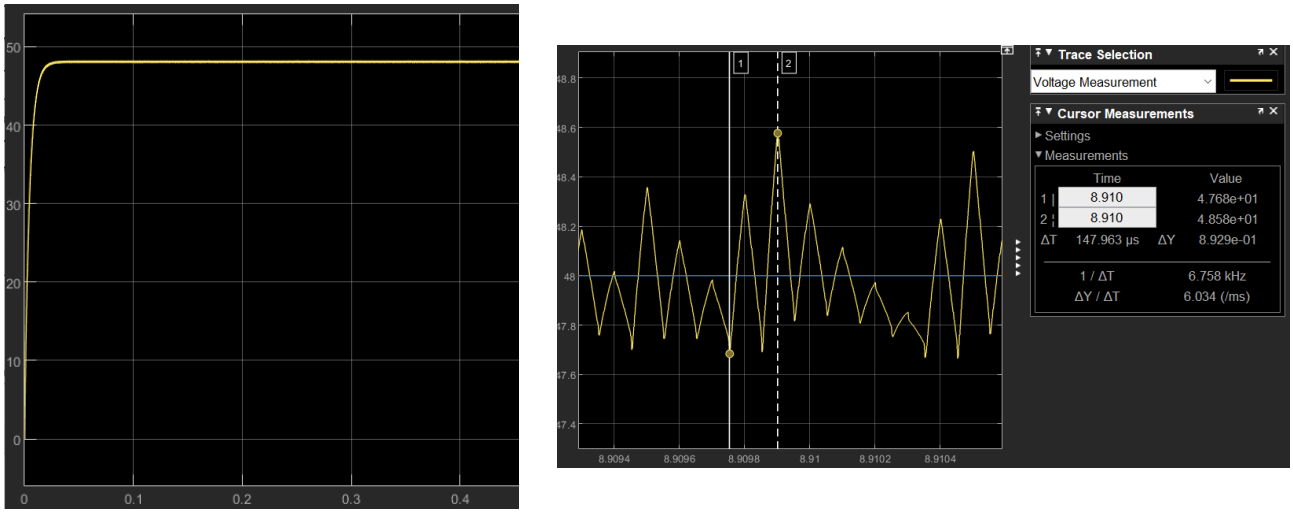


Figure 22: Voltage variation and ripple estimation

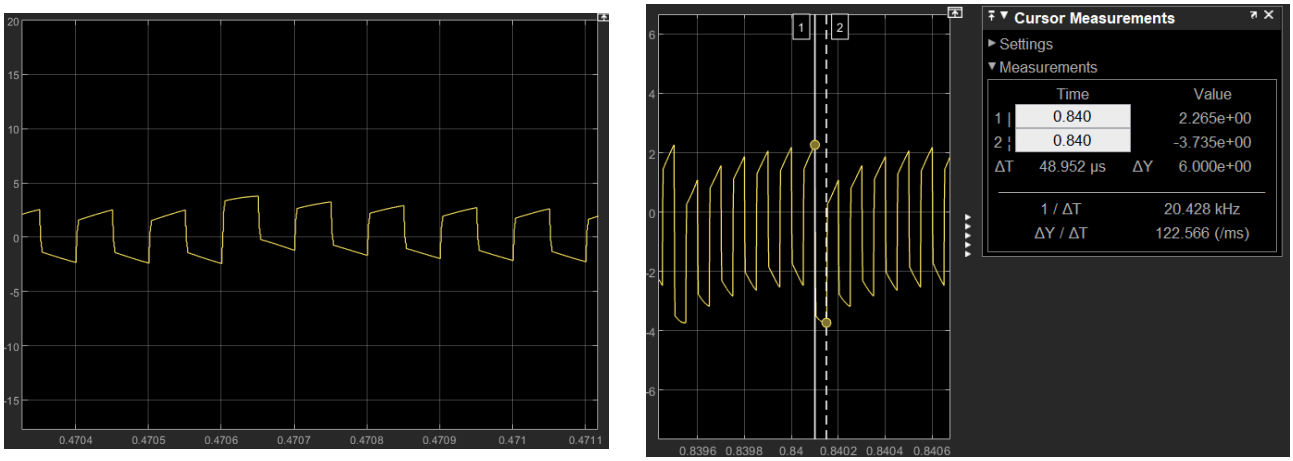


Figure 23: Current waveform and ripple estimation

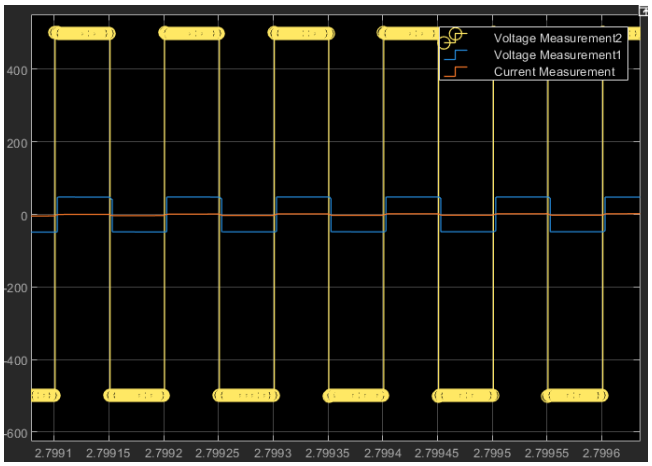


Figure 24: Transformer Voltages and current

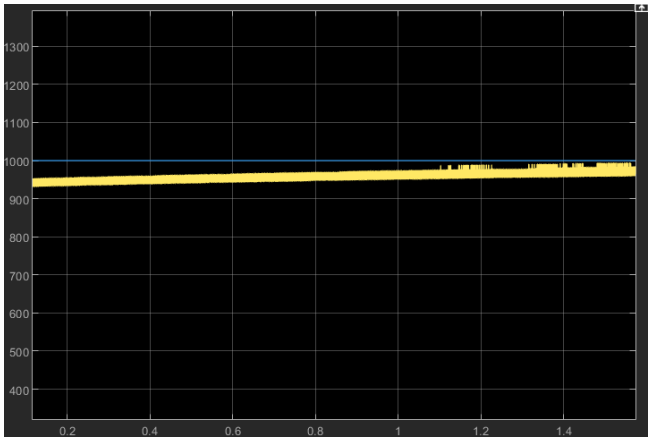


Figure 25: Power out

6 Conclusion

- **Objective**: Designed a 1kW bidirectional dual-active bridge converter for automotive auxiliaries, converting 500V DC to isolated 12V and 48V outputs
- **Key Parameters:**
  - Transformer turns ratio: 10.42:1 (48V) & 41.67:1 (12V)
  - Leakage inductance: 500 μH
  - Output capacitance: 1,500 μF
  - Ripple targets: <10% current ripple & <0.5% voltage ripple
- **Component Selection:**
  - SiC MOSFETs: Optimized for low conduction/switching losses
  - Planar transformer: Compact design with controlled leakage
- **Performance:**
  - Efficiency: 97.5% (simulated) & 98.3% (analytical)
  - Stable operation: Output ripple & transient response within specs
- **Validation & Applications:**
  - Verified the **first iteration** via MATLAB/Simulink simulations & analytical calculations
- Despite not being able to simulate the Centre-tapped output to get 12V owing to configuration confusions, the basement required to build the proposed converter has clearly been laid and with little guidance, the Centre-tapped multi-output phase shifted full bridge converter can be designed as proposed.